



Gwanwoo Ju

E-mail: gwanwoo7835@jbnu.ac.kr (gwanwoo7835@gmail.com)

Mobile: +82 10-6275-7835

Address: Jeonju, Republic of Korea

Language | Korean, English

Education

03/2019 – 02/2025

- Bachelor's degree, Department of Mineral Resources and Energy Engineering, Jeonbuk National University, South Korea

03/2025 – Current

- Integrated Ph. D Course, Department of Climate, Environment and Energy, Jeonbuk National University, South Korea

Working Experiment

08/2025– 03/2026

- Visiting Student Researcher (VSR) in Department of Geomatics Engineering, University of Calgary

Area

- Reservoir simulation study for conventional & unconventional reservoir
- CCS reservoir simulation study
- Optimization study using various algorithms
- Machine learning

Research Projects

04/2024 – 09/2024

- Canada-South Korea Climate Action Partnership: Emission Reduction Paths and Effects Evaluation

08/2024 – 07/2026

- Development of Screening Criteria for Enhanced CO2 Injection and Optimal Injection Well Patterns into the aquifer

03/2025 – 12/2026

- Global Talent Development Program for Innovative Smart Carbon Capture and Storage Operations through Artificial Intelligence and Digital Twin Integration

Training

- Underground Carbon Storage Modeling Using STOMP CO2 Simulator (KIGAM, 2023)
- Underground Hydrogen Storage Simulation & CO2 EOR Modeling (CMG, 2024)
- Eclipse Black Oil Reservoir Simulation (SLB, 2024)
- Underground CO2 Storage Modelling & Geomechanics Modelling (CMG, 2024)
- Eclipse Compositional Simulation (SLB, 2025)
- Introduction to CO2 Flow Assurance in transportation and injection(SLB, 2025)
- Reservoir Simulation using Petrel RE and INTERSECT (SLB, 2026)

Conference Proceedings

- [Ju, G.](#)

Awards

- [Ju, G.](#), Excellence Award, The 1st AI Applications in Hydrogen & Gas Industry Best Practice Contest, Hosted by the Korean Institute of Gas (KIGAS), May 28, 2026. (*Title: Multi-Objective Optimization Framework for Geologic Carbon Storage*)

Publications

- Lee, Y., [Ju, G.](#), Kim, M., Kim, J., Lee, Y.*. Physics-Informed Machine Learning for Rapid Assessment of CO2 Injectivity and Plume Migration: From Surrogate Modeling to Field-Scale Application (*Under review*)
- Lee, Y., Dehghan, M. J., Kim, J. W., Ali, M. Y., [Ju, G.](#), Oh, J. Wavenumber Correlation of Gravity and Magnetic Anomalies for Constraining Salt-Basement Structure in Offshore Abu Dhabi (*Under review*)

Patent

- [Ju, G.](#), Lee, Y. Apparatus and Method for Evaluating CO2 Geological Storage Feasibility in Aquifers (*Patent Pending*)